



Fig. 4: Atmospheric losses of the electromagnetic radiation

ing new nonlinear phenomena and for pump-probe studies of dynamical properties of materials. Table-top systems using optical rectification of femtosecond lasers either at high repetition rates or high peak power are routinely available.

Advanced communications. The electromagnetic spectrum has three important windows: radio, optical and infrared. Radiations with these frequencies in the windows can propagate without being absorbed by natural materials or the atmosphere. The radio window has been widely used for radio, television and radar, and the use of optical window is now underway. Yet, very little achievement has been made in the use of THz near-infrared windows.

Frequencies higher than microwave offer many benefits for the communication technology:

1. Broad bandwidth for high-data-rate information transmission
2. High-speed information data transmission through telecommunication networks
3. High directivity and spatial resolution
4. Low transmission power
5. Low probability of interference/interception
6. Small antenna and equipment size
7. No multipath fading

The obvious disadvantages of communications at THz arise from the strong absorption through the atmosphere caused by water vapour as well

as the low efficiency and relatively low power available from existing sources (see Fig. 4). Despite these disadvantages, many communication applications are possible for this frequency range.

For satellite-to-satellite communications, atmospheric absorption is not a problem, except for paths that graze the earth's atmosphere. Here the advantages of using THz technology are the larger bandwidth and therefore high data-transmission rate compared to microwave communication. Indoor wireless communications with THz may provide multiple data channels with gigabit-per-second or greater capacity.

One problem encountered in optical communications is attenuation caused by scattering and absorption by clouds, rain, dust, etc. In the Rayleigh scattering regime, the scattering cross-section increases with shorter wavelengths as the fourth power of the inverse wavelength. Thus THz radiations can be used for short distances if heavy particulates (smoke and dust) are present in the air.

The great successes in electromagnetic engineering have been possible in this part because the wavelengths are either much longer than relevant component sizes or much smaller. THz radiation can transmit through materials that we normally think of as opaque: thick smoke, cloud, fog, dust or even the dry sand of a sandstorm. While this neat trick can also be accomplished at some lower frequencies (millimetre waves, for instance), at THz frequen-

cies there is a crucial advantage.

Many other kinds of THz sensor and communications systems could be possible, each with its own unique and important advantages: For instance, communication systems with enormous information bandwidths that can propagate through a dark cloud and secure communications that cannot propagate to where an enemy might easily detect.

THz electro-optic transceiver is used for THz sensing, ranging and imaging applications. It emits and receives THz signals by using a ZnTe crystal with femtosecond optical pulses. It is based on the reciprocal behaviour of optical rectification and electro-optic effect in the non-linear optical crystal. With optical excitation and sampling of an ultrafast laser, the electro-optic THz transceiver alternately transmits pulsed electromagnetic radiation and receives the return signal from the same crystal.

Anti-terror devices. Recent work shows that THz technologies are promising for the standoff detection and identification of explosives targets and for defense and security applications. The spectral domain from 0.1 to 10 THz hosts low-frequency crystalline lattice vibrations (phonon modes), hydrogen-bonding stretches and other intermolecular vibrations of molecules in many chemical and biological materials, including explosives and related compounds (ERCs), and drugs.

Many polar gases also have distinctive spectroscopic fingerprints characterising these THz vibrational modes and providing information that is not available in other parts of the electromagnetic spectrum. THz waves have low photon energies (4 meV for 1 THz, one million times weaker than X-ray photons) and don't cause harmful photoionisation in biological tissues. As a potential sensing and imaging modality, THz radiation is considered safe for the operators and targets.

THz radiation can penetrate through many commonly used non-polar dielectric materials, such as